

Generative AI 101

Take-Home Reference Guide

A complete synthesis of all topics covered in today's training.

I. What Is Generative AI?

Traditional AI	Generative AI
Systems that perceive their environment, reason through problems, and take actions. • Pattern recognition • Decision making • Problem solving • Data analysis	A specialized subset that doesn't just analyze — it generates original outputs based on learned patterns. • Content creation • Text generation • Image synthesis • Code writing

The Four Layers of AI Evolution

Each layer builds on the one before it:

- **Artificial Intelligence** — The foundational layer: all systems that simulate human intelligence.
- **Machine Learning** — Systems that learn from data without explicit programming.
- **Deep Learning** — Neural networks with multiple layers, mimicking how the brain processes information.
- **Generative AI** — Creates new, original content by understanding patterns from training data.

Key Mindset Shift

Generative AI is an advanced probability engine — it predicts tokens, not concepts. It is not smart; it is fast. Your role shifts from primary creator to quality assurance. Hallucinations are a feature of how the system works, not a bug.

How It Works: Tokenization

The model receives your prompt → splits it into tokens (words, parts of words, or characters) → predicts the next most likely token → repeats until done.

Pro Tip

Understanding tokenization explains unexpected outputs. The model is completing a probability sequence, not thinking through your question. Precision in prompts directly controls the quality of the token predictions.

II. Prompting Fundamentals

Input quality = output quality. This is not a metaphor. Every prompt is an instruction set. Treat it like one.

Prompting Techniques

Zero-Shot Prompting

What it is: A direct, single instruction with no examples or context.

Best for: General knowledge, quick answers, initial brainstorming, common professional topics.

Example: "Explain the importance of learning and development for HR professionals in modern organizations."

Few-Shot Prompting

What it is: Providing context, examples, or prior information to shape the response.

Best for: ~80% of your prompting should be this. Specificity always helps.

Example: "Plan a vacation to Barcelona. I've been to Madrid and Lisbon before and would like to see new landscapes and try highly rated local restaurants. I'd also like to visit a nature reserve."

Retrieval Augmented Generation (RAG)

What it is: Grounding the AI response by providing a specific source URL, document, or data.

Process: Query & Intent → Knowledge Retrieval → Context Augmentation → Grounded Generation.

Example: "Explain the importance of L&D for HR using this URL as supporting stats: <https://www.aihr.com/blog/learning-and-development-statistics/>"

Advanced Techniques

Chain of Thought

What it is: Asking the AI to walk through each logical step in its reasoning before reaching a conclusion.

Best for: Sequential planning, step-by-step checklists, process documentation.

Example: "Use chain of thought to provide a step-by-step checklist in timeline format on what I should do, pack, and prepare for my trip to [PLACE]."

Tree of Thought

What it is: Generating multiple distinct reasoning paths or options simultaneously.

Best for: Comparing alternatives, exploring options, strategic decisions.

Three questions to ask: What's your target? How many outputs? What's the format?

Example: "Use tree of thought to provide 3 potential alternative strategies for improving employee retention, each with different assumptions and tradeoffs."

Metaprompting

What it is: Asking AI to identify what you haven't considered, improve your prompt, or stress-test your thinking.

Best for: Project planning gaps, risk assessment, stakeholder analysis, blind spot discovery.

Example: "What are we not thinking of for our upcoming product launch that could cause problems or missed opportunities?"

Prompt Structuring Tools

Two simple tools that meaningfully improve output quality:

Hashtags (#)

Use hashtags to categorize and structure prompt components. This helps the AI parse your intent more accurately.

#Role: Senior L&D Specialist

#Task: Create onboarding plan

#Audience: Remote employees

#Format: Weekly breakdown

Brackets ([])

Use brackets to mark placeholders, optional parameters, or variables for repeatable templates.

"As a [Role], draft a [Content Type] about [Topic] for [Audience]. Tone should be [Tone]."

Combine both for maximum reusability.

UPSCALE

III. Adjusting Parameters

Most conversational AI tools let you control these settings. Understanding them helps you get consistent, intentional outputs.

Parameter	What It Does	When to Adjust
Temperature	Controls randomness/creativity. 0 = deterministic, identical outputs. 1.0 = maximum variation.	Low (0–0.3) for factual, structured outputs. Higher (0.7–1.0) for creative writing, brainstorming.
Top P (Nucleus Sampling)	Limits the pool of word choices to the most probable set. More focused than temperature alone.	Keep around 0.9 for balanced output. Lower (0.3) with high temperature to maintain coherence.
Presence Penalty	Penalizes tokens that have already appeared, regardless of how many times.	Increase for creative tasks where word variety matters. Lower for repetitive, structured documents.
Frequency Penalty	Penalizes tokens based on how many times they've already been used.	Increase for poetry, creative writing. Lower for technical, structured, or compliance-sensitive content.

UPSCALE

IV. Prompt Frameworks

Frameworks give you consistency, clarity, and repeatability. Once mastered, you apply them automatically.

RTF: Role – Task – Format

Role: Define who the AI should act as (HR manager, L&D specialist, legal reviewer, etc.).

Task: Specify what you want accomplished: create, analyze, summarize, recommend, evaluate.

Format: Describe the output structure: bullet points, email, table, checklist, slide outline.

Example: "You are an experienced L&D specialist. Create a 30-day onboarding plan for remote employees. Present it as a weekly breakdown with specific activities and checkpoints."

BAB: Before – After – Bridge

Before: Describe the current challenging situation or pain point.

After: Describe the desired future state.

Bridge: Specify the solution that connects the two.

Example: "Before: Employee struggles with time management. After: Confident, organized, productive team member. Bridge: Build a structured 60-day development plan with weekly coaching touchpoints."

RODES: Role – Objective – Details – End – Sense Check

Role: Define the persona for tone and perspective.

Objective: State the main goal of the interaction.

Details: Provide all context, constraints, and specific information.

End Goal: Specify desired output format and delivery requirements.

Sense Check: Instruct the AI to review its own output for accuracy and coherence before finalizing.

Onion Framework: Peel Away Overwhelm

Purpose: Reduce cognitive overload by categorizing all active items and forcing deliberate removal of low-value tasks.

Process: Input all active tasks → Categorize into Core / Important / Surface Noise / Remove → Identify top 3 priorities for the next 90 minutes.

Peel Rule: Recommend removing 20–30% of captured items immediately.

Philosophy: Forward progress doesn't require holding onto everything. Protective clutter is not productivity.

Gravity Framework: Stress-Test Your Ideas

Purpose: Expose weak reasoning, flawed assumptions, blind spots, and execution risks before committing resources.

Role: Instruct AI to act as intellectual gravity — prioritize truth over encouragement.

Output: Core vulnerabilities ranked by severity; faulty assumptions; overlooked risks; what would cause this to fail; what must be true for this to succeed.

Philosophy: Strong ideas survive pressure. Weak ideas rely on momentum.

Build Your Own Framework

Identify a recurring task → Define the essential inputs the AI needs → Sequence and name it → Test and refine.
Include both hashtags and brackets for maximum reusability.

UPSCALE

V. System Prompts

A system prompt is a persistent set of instructions applied to every interaction in a session. Unlike regular prompts, it shapes the AI's default behavior, tone, and focus automatically.

Why Use System Prompts?

- Eliminates the need to repeat context and preferences in every prompt.
- Maintains consistent tone, expertise level, and format across extended conversations.
- Keeps AI focused on your specific role and organizational context.

How to Build One

Step 1 — Define Role & Voice — Who should the AI be? What communication style? (e.g., "You are a senior HR consultant who communicates in plain language.")

Step 2 — Set Clear Boundaries — What should the AI never do? Include compliance, confidentiality, or content restrictions.

Step 3 — Clarify Style & Format — Response length preference, formal vs. conversational, any industry-specific terminology.

Step 4 — Include Persistence Rules — How should the AI handle unclear requests? Should it ask clarifying questions or make reasonable assumptions?

Pro Tip

Start with 2–3 key instructions and refine based on actual usage. A system prompt doesn't need to be long to be effective. More instructions can introduce conflicts.

Example System Prompt

You are a policy editor specializing in HR documentation. Always rewrite content to be clear, compliant, and employee-friendly. Use bullet points for complex procedures. Ask clarifying questions if policy details are ambiguous. Maintain professional tone while ensuring accessibility for all reading levels.

VI. AI Agents

Agents shift the model from something you direct step-by-step to something that pursues goals autonomously — planning, using tools, iterating, and self-correcting.

The Progression

Stage	Description
Prompting	Direct, single-turn instructions for immediate response.
Structured Prompting	Frameworks and advanced techniques for consistent, high-quality outputs.
Tool Use	LLMs accessing external APIs, databases, or functions to take specific actions.
AI Agents	Autonomous systems capable of planning, executing, and refining multi-step tasks without constant human direction.

What Makes an Agent Different

- **Goal Orientation** — Agents define and pursue high-level objectives, breaking them into sub-tasks.
- **Planning Loop** — They continuously strategize, adapt plans, and self-correct based on feedback.
- **Tool Access** — Agents can use external APIs, databases, and software to gather information and take action.
- **Memory & Context** — They maintain persistent state, learning from prior interactions.
- **Iteration & Refinement** — They observe outcomes, evaluate against goals, and iterate.

When to Use an Agent vs. a Prompt

Use an Agent When: • Task requires multi-step execution with dynamic sequencing. • You need to integrate external data sources or tools. • Conditional logic and autonomous decision-making are needed. • Goals are high-level and require autonomous breakdown. • Ongoing monitoring and adaptive adjustment are required.	Skip the Agent When: • Single, straightforward transformation task. • Deterministic workflow with fixed, predictable steps. • One-off content generation or simple summarization. • A clear, unchanging checklist covers the task completely.
--	--

The Rule

If a checklist works perfectly, you don't need an agent. Only add agents when tasks demand autonomous planning, adaptation, and tool use beyond simple instruction following.

Agent Risks & Controls

Key Risks • Hallucinated tool calls — misinterpreted instructions. • Looping failures — stuck in unproductive cycles. • Security exposure — expanded access surface. • Cost creep — unmonitored API usage.	Practical Controls • Human-in-the-loop checkpoints before critical actions. • Usage caps on API calls and execution time. • Logging and monitoring of all agent decisions. • Clearly defined goal constraints and operational scope.
---	---

VII. Effective Automation

Automation compounds. Small wins stack into operational leverage. But poor automation choices cost more than they save. Use a deliberate process.

The Automation Journey

- **1. Assess the Need** — Identify top 5 weekly tasks by time consumed. Classify them: mechanical (automatable) vs. judgment-required. Flag scalability bottlenecks.
- **2. Quantify the Pain** — Estimate total staff-hours lost monthly. Calculate cost of errors and delays. Document compliance and audit risks with financial exposure.
- **3. Prioritize Initiatives** — Use the Priority Matrix (below) to rank candidates by feasibility and impact.
- **4. Align Stakeholders** — Identify champions at each org level. Host 30-minute working sessions per group. Document objections. Define shared success metrics.
- **5. Plan for Resilience** — Build a PACE Plan (see below) before deploying anything.

Priority Matrix

Score each automation candidate 1–5 on both axes:

Feasibility (1–5)	Impact (1–5)
Technical readiness, budget availability, timeline, team capability.	Time saved, cost reduced, risk mitigated, revenue enabled.
High Feasibility + High Impact QUICK WINS — Do these first.	Low Feasibility + High Impact STRATEGIC PROJECTS — Plan and invest.
High Feasibility + Low Impact FILL-INS — Only if resources allow.	Low Feasibility + Low Impact AVOID — Not worth the effort.

The PACE Plan

Pre-define your response to every level of system failure before you deploy. Don't design under pressure.

	What It Means	Key Requirements
PRIMARY	The ideal, fully automated workflow under normal conditions.	Define core solution, success metrics, alert thresholds, and ownership.
ALTERNATE	A pre-configured backup system or process ready to activate.	Must be pre-built, not planned. Define hand-off triggers and transition protocol.
CONTINGENCY	A simplified, semi-manual process for when both P and A fail.	Design before any incident. Small team can run essential functions manually.
EMERGENCY	Fully manual, bare-minimum survival operations.	Document last-resort steps, notification chain, max tolerable downtime threshold.

Quick Reference Cheat Sheet

Cut this page out and keep it at your desk.

Technique	When to Use	Quick Trigger
Zero-Shot	Quick answers, common topics, initial brainstorm	"Explain [X]"
Few-Shot	Most professional tasks — add context and examples	"Here's what I know... now [task]"
RAG	When you need facts grounded in a source	"Using this URL/doc as context..."
Chain of Thought	Complex sequential tasks, checklists	"Step by step..."
Tree of Thought	Comparing options, strategic decisions	"Give me 3 paths to..."
Metaprompting	Blind spot discovery, risk analysis	"What am I not considering?"
RTF	Any professional content creation	Role + Task + Format
BAB	Change initiatives, coaching plans	Before + After + Bridge
RODES	Detailed, multi-part deliverables	Role + Objective + Details + End + Sense Check
Onion	Task overload, prioritization	"Apply the Onion framework to my tasks..."
Gravity	Idea validation, risk exposure	"Stress-test this idea..."
System Prompt	Repeated workflows, consistent persona	Set once in Settings or start of session

The Most Important Rule

Verify AI outputs before using them. You are now the quality assurance layer. AI is fast and often wrong. Your judgment is the last gate before a hallucination becomes a mistake.

Your Next Step

Pick one task you do repeatedly every week. Build a reusable prompt template using RTF or RODES. Test it. Refine it. That one template, used consistently, will save you hours this month.